Noise and Outliers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected summary and data-audit record.

Count repeated values, sort for two median cases, challenge unsupported deletion and report missing summaries.

Keep one habit: Keep raw observations, changed-case status, units, calculations and source decisions visible.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.
- Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Overfitting Intuition

Decision Trees

Training and Testing

W1 · Compute a mean with repeated observations

> The denominator counts observations rather than distinct values.

Your evidence record

- Summary reference: mean=sum/count, counting repeated observations separately. For a nonempty sorted odd list take its middle; for an even list average the two middle values. Empty data has no measured mean or median here. Worked baseline durations are2,3,3,4,18 minutes;18 is not confirmed wrong. Keep hypothetical18→8 calculations separate from the authorised18→3 log scenario.

- Find the **mean** of 4, 4, 6, 6 and 10.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Compute odd and even medians

> Sort each supplied list before choosing the middle.

Your evidence record

- Summary reference: mean=sum/count, counting repeated observations separately. For a nonempty sorted odd list take its middle; for an even list average the two middle values. Empty data has no measured mean or median here. Worked baseline durations are2,3,3,4,18 minutes;18 is not confirmed wrong. Keep hypothetical18→8 calculations separate from the authorised18→3 log scenario.

- Find **median**s of [9, 1, 3] and [8, 2, 4, 6].

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Challenge deletion based on the mean

> State the source evidence and collection rule you need.

Your evidence record

- Summary reference: mean=sum/count, counting repeated observations separately. For a nonempty sorted odd list take its middle; for an even list average the two middle values. Empty data has no measured mean or median here. Worked baseline durations are2,3,3,4,18 minutes;18 is not confirmed wrong. Keep hypothetical18→8 calculations separate from the authorised18→3 log scenario.

- A learner deletes 18 because it is above the **mean**. What **evidence** is needed before correcting or excluding it?

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Report an unavailable mean honestly

> Do not replace absence of observations with a measured zero.

Your evidence record

- Summary reference: mean=sum/count, counting repeated observations separately. For a nonempty sorted odd list take its middle; for an even list average the two middle values. Empty data has no measured mean or median here. Worked baseline durations are2,3,3,4,18 minutes;18 is not confirmed wrong. Keep hypothetical18→8 calculations separate from the authorised18→3 log scenario.

- A dataset is empty. Is reporting a **mean** of 0 justified?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Compare summaries independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Treat the independent list as durations in minutes; the requested replacement is a what-if comparison, not an evidence-backed correction of actual data.

For 1, 2, 2, 3 and 12, calculate **mean** and **median**. Then replace 12 with 2 and recalculate. Explain the change.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Compare summaries independently

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Keep legitimate unusual books so the recommender can serve unusual readers.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Compute two **median**s for **Mean median** and **unusual value**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Mean median** and **unusual value**s. Proposed explanation: "A successful run proves the conclusion". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Page counts are 72,80,96,960. The last book may be an anthology. Should a cleaning script delete it automatically? Give a defensible investigation and document the outcome.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Recorded lengths are 90,95,100,105,900 pages.

Is 900 automatically wrong?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Page counts are 40, 42, 43 and 900. Should 900 be deleted immediately? Give two explanations to **investigate**.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 17 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Page counts are 72,80,96,960. The last book may be an anthology. Should a cleaning script delete it automatically? Give a defensible investigation and document the outcome.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

For 1, 2, 2, 3 and 12, calculate **mean** and **median**. Then replace 12 with 2 and recalculate. Explain the change.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

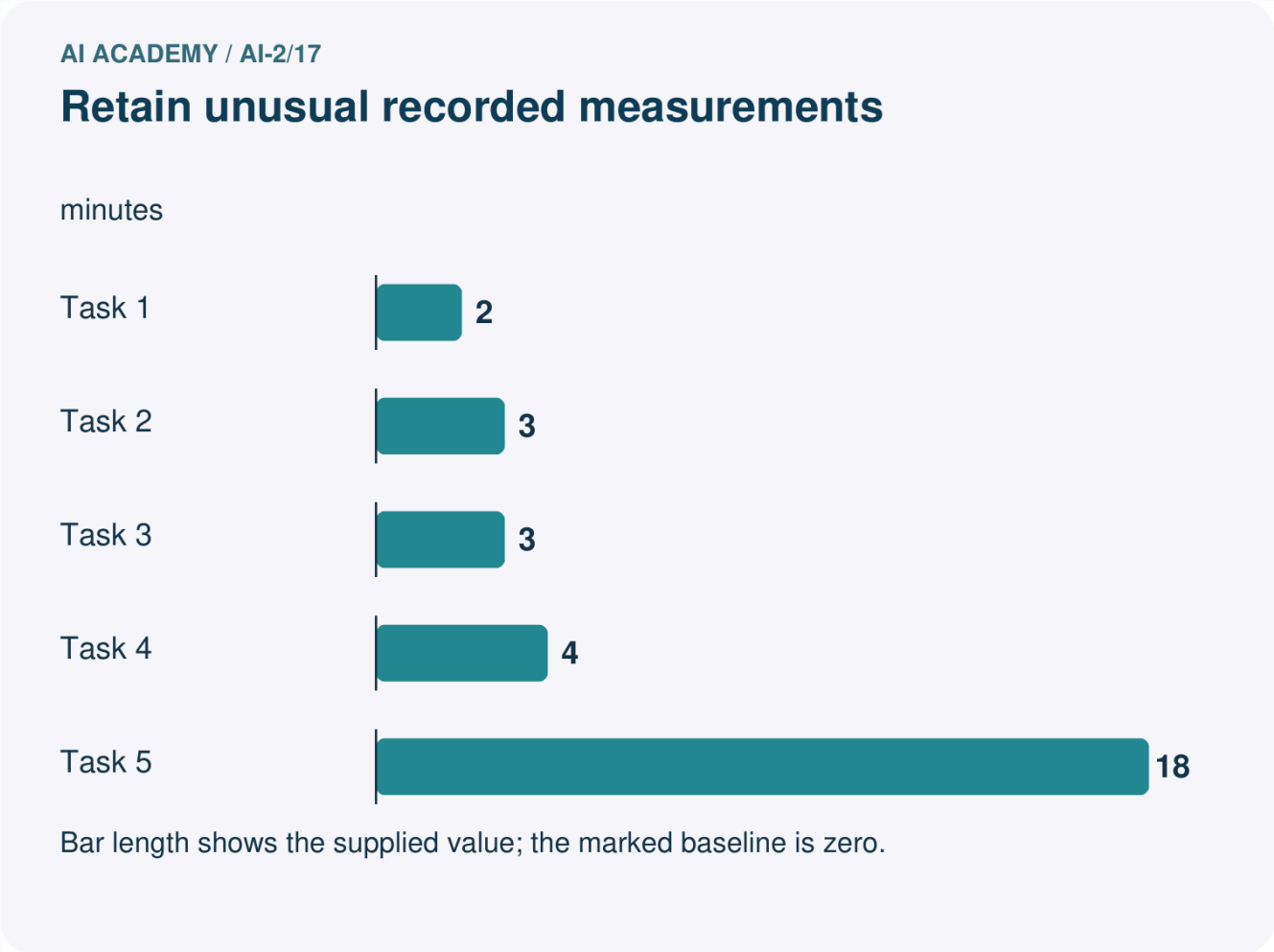
> Keep the explanation and the support you used.

Use this when helpful

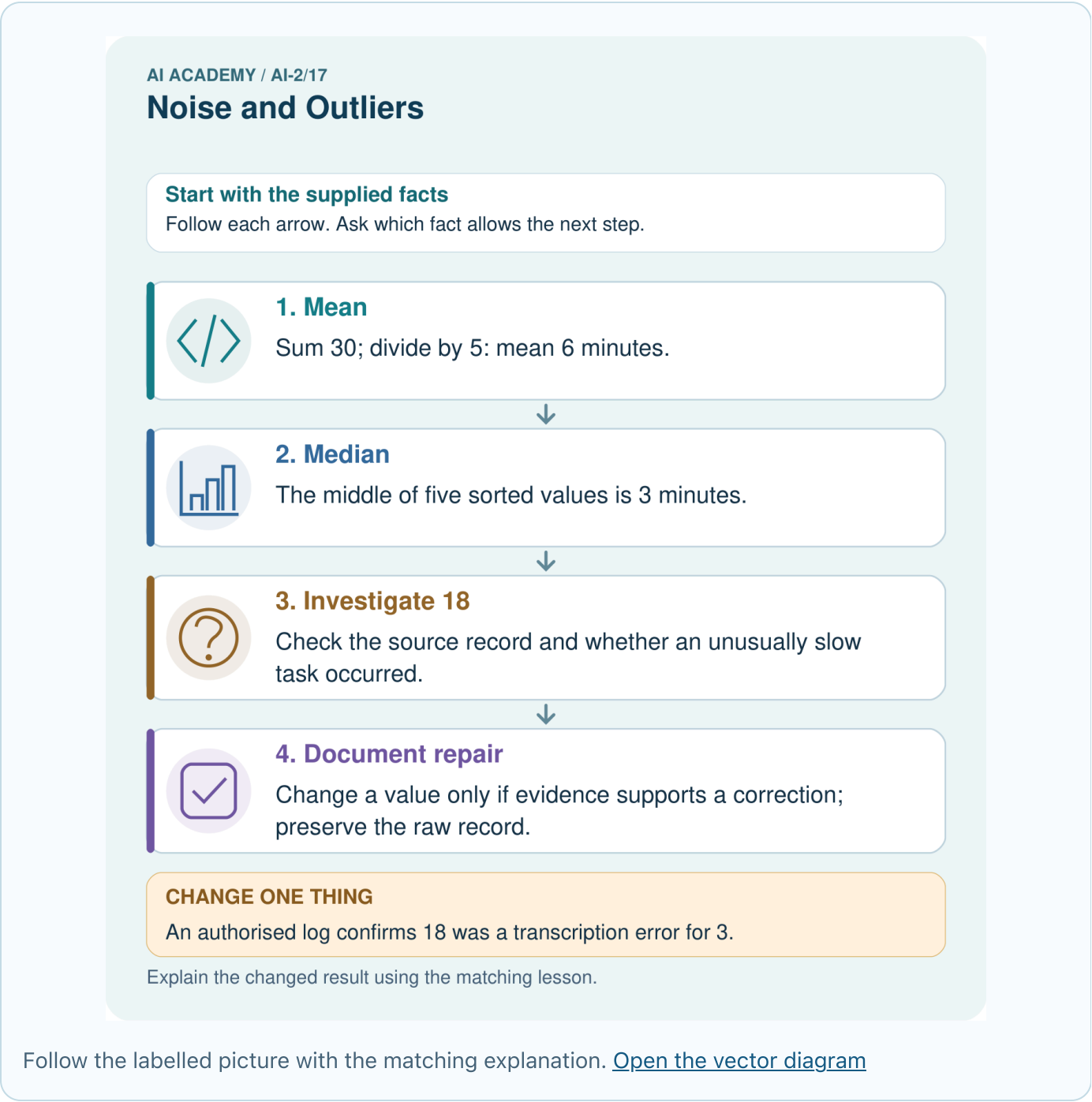
A fictional library records five book-**process**ing times in minutes.

Sorted times: 2,3,3,4,18.

All are recorded measurements; no **evidence** yet shows 18 is a mistake.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: **Investigate** unusual recommender features and show **evidence**-backed repairs instead of removing inconvenient records.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Investigate unusual recommender features and show **evidence**-backed repairs instead of removing inconvenient records.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional library records five book-**process**ing times in minutes.
- Which labelled step matches this detail? Change a value only if **evidence** supports a correction; preserve the raw record.
- 1. **Median**
- 2. **Investigate** 18
- 3. Document repair
- 4. **Mean**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Document repair Change a value only if **evidence** supports a correction; preserve the raw record. The **mean** uses every value and is pulled upward by 18.

The **median** identifies the middle recorded time and stays at 3. Neither statistic declares the unusual **observation** invalid.

Deleting it simply because it changes the **mean** would hide potentially real slow work and change the question being answered.

Changed-case question

- Changed-case practice: An authorised log confirms 18 was a transcription error for 3. Which conclusion is supported by the supplied facts?
- 1. Corrected **mean** is 3 minutes and **median** is 3.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Corrected **mean** is 3 minutes and **median** is 3. Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Repair and save your work

Review Over**fitting** Intuition (ai-2/16). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does zero **training** error justify choosing Complex in the original record?.

Return to this week and explain what you changed.

Review Decision Trees (ai-2/14). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Why does changing colour alone not change this tree's **prediction**?.

Return to this week and explain what you changed.

Review **Training** and Testing (ai-2/11). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan counts?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

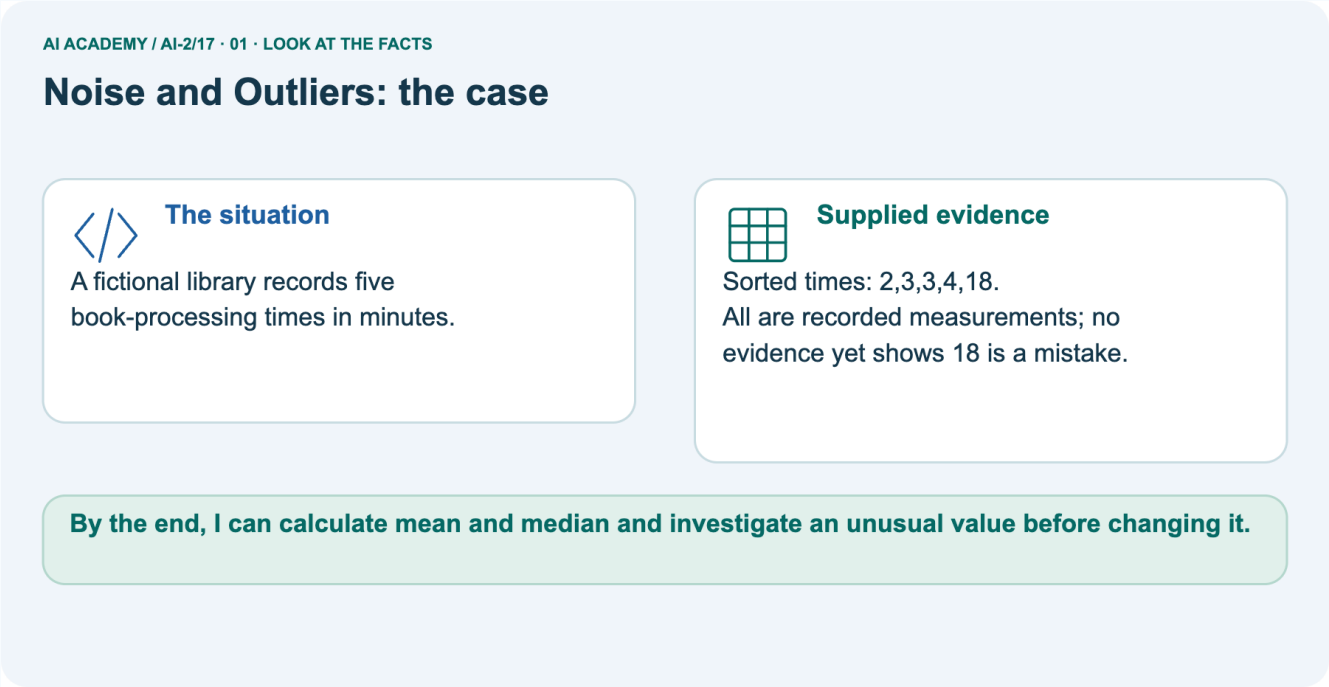
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Noise and Outliers: the case



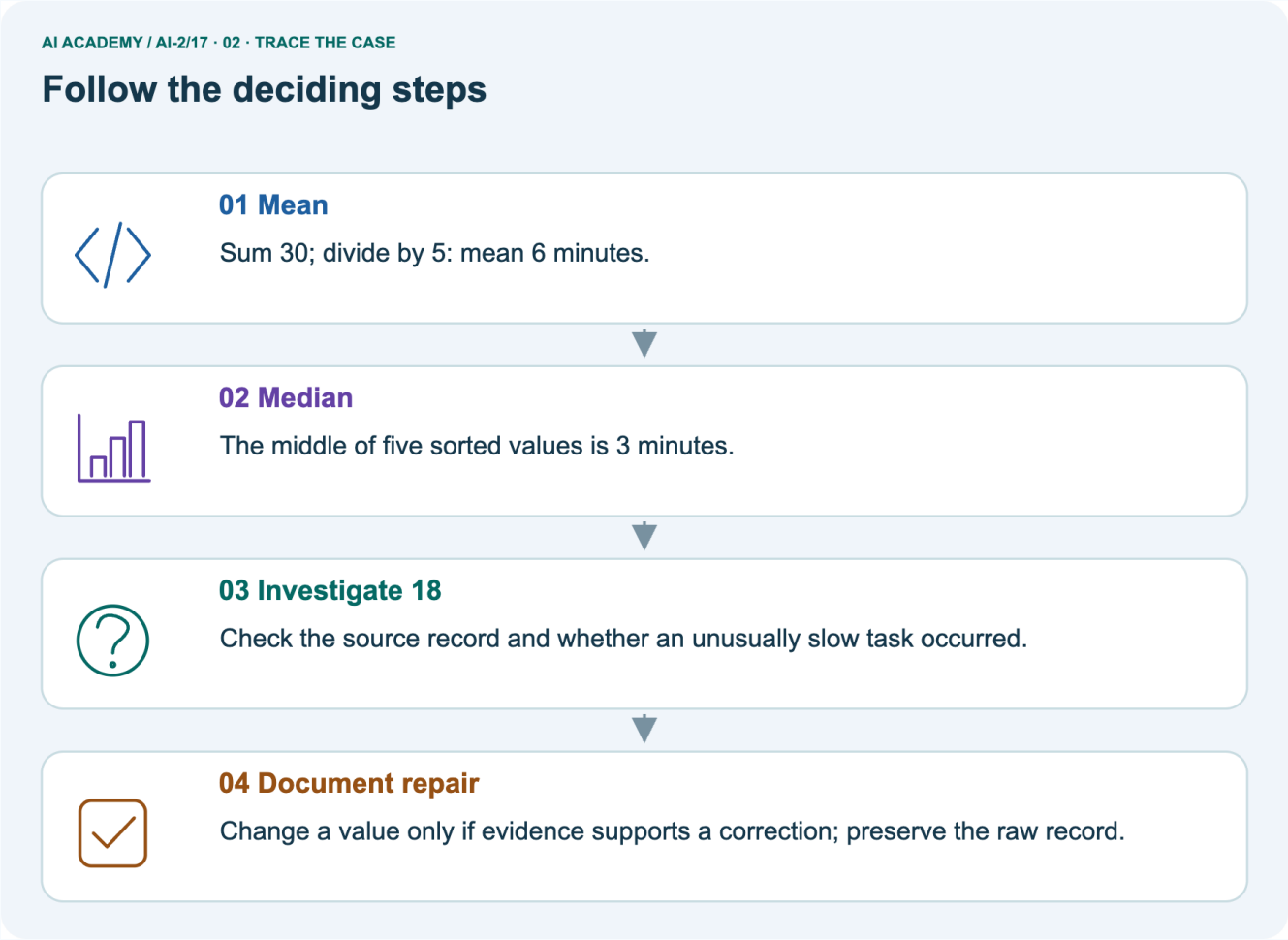
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional library records five book-**process**ing times in minutes. Supplied facts: Sorted times: 2,3,3,4,18. All are recorded measurements; no **evidence** yet shows 18 is a mistake.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



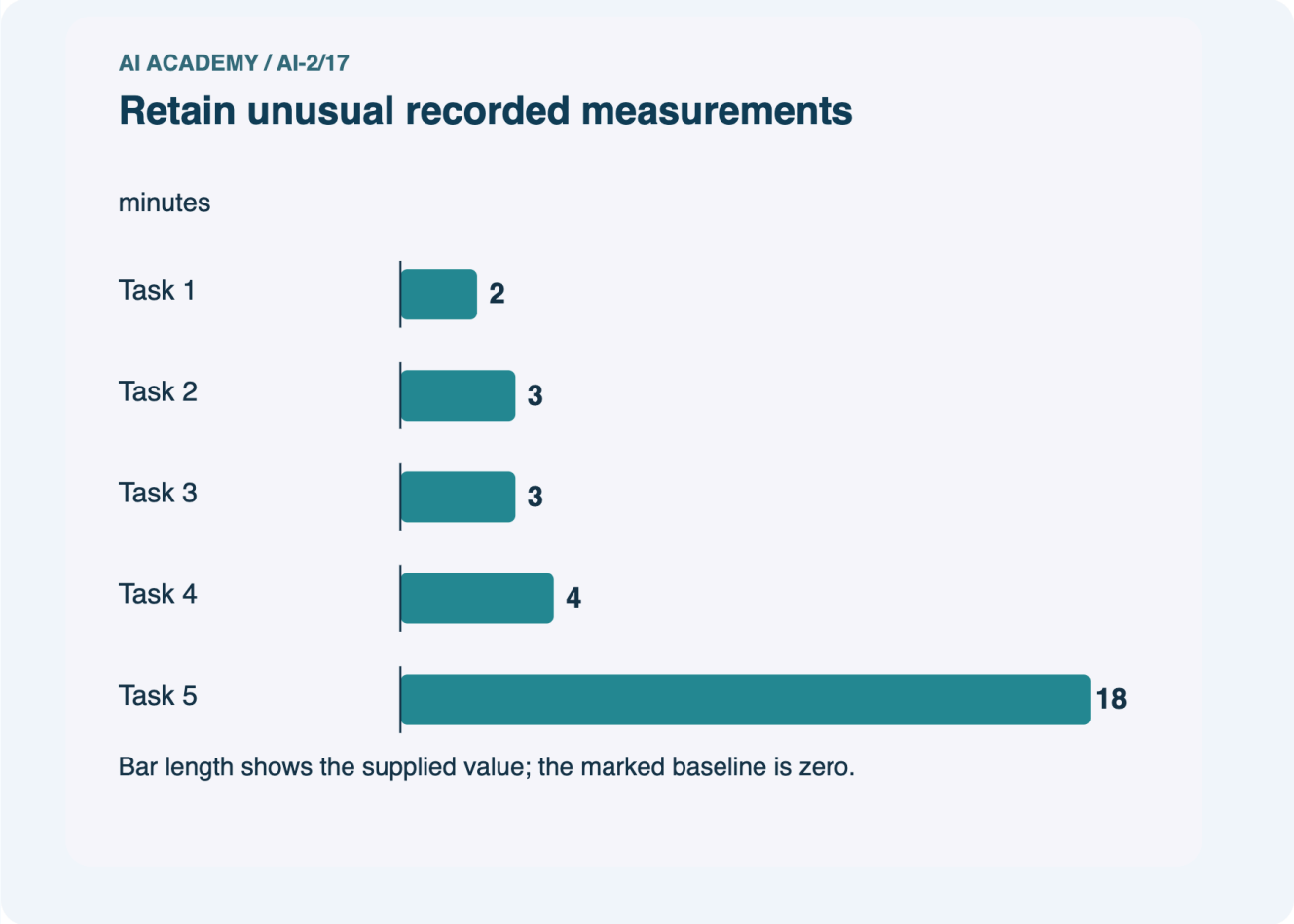
Follow the labelled picture with the matching explanation. Open the vector diagram

Mean: Sum 30; divide by 5: **mean** 6 minutes. → **Median**: The middle of five sorted values is 3 minutes. → **Investigate** 18: Check the source record and whether an unusually slow task occurred. → Document repair: Change a value only if **evidence** supports a correction; preserve the raw record.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Retain unusual recorded measurements



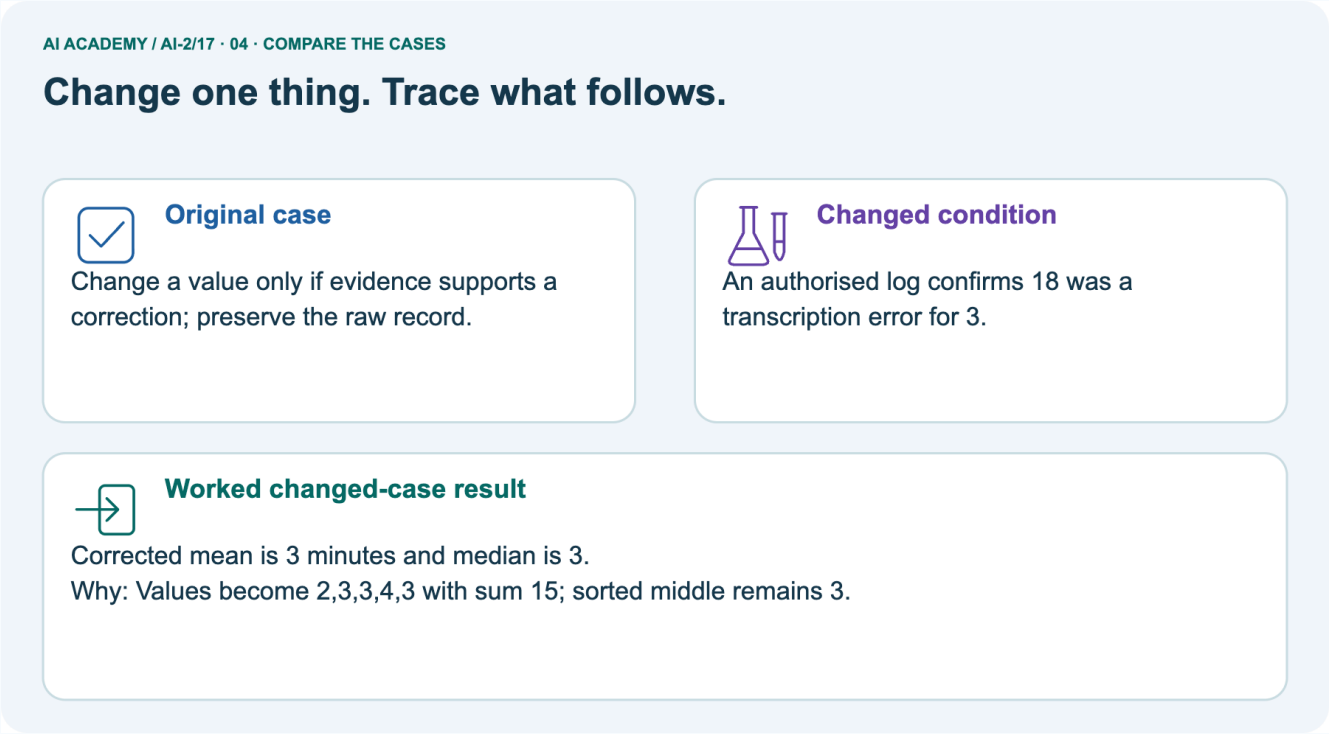
Follow the labelled picture with the matching explanation. Open the vector diagram

Retain unusual recorded measurements. Units: minutes. Task 1: 2. Task 2: 3. Task 3: 3. Task 4: 4. Task 5: 18.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

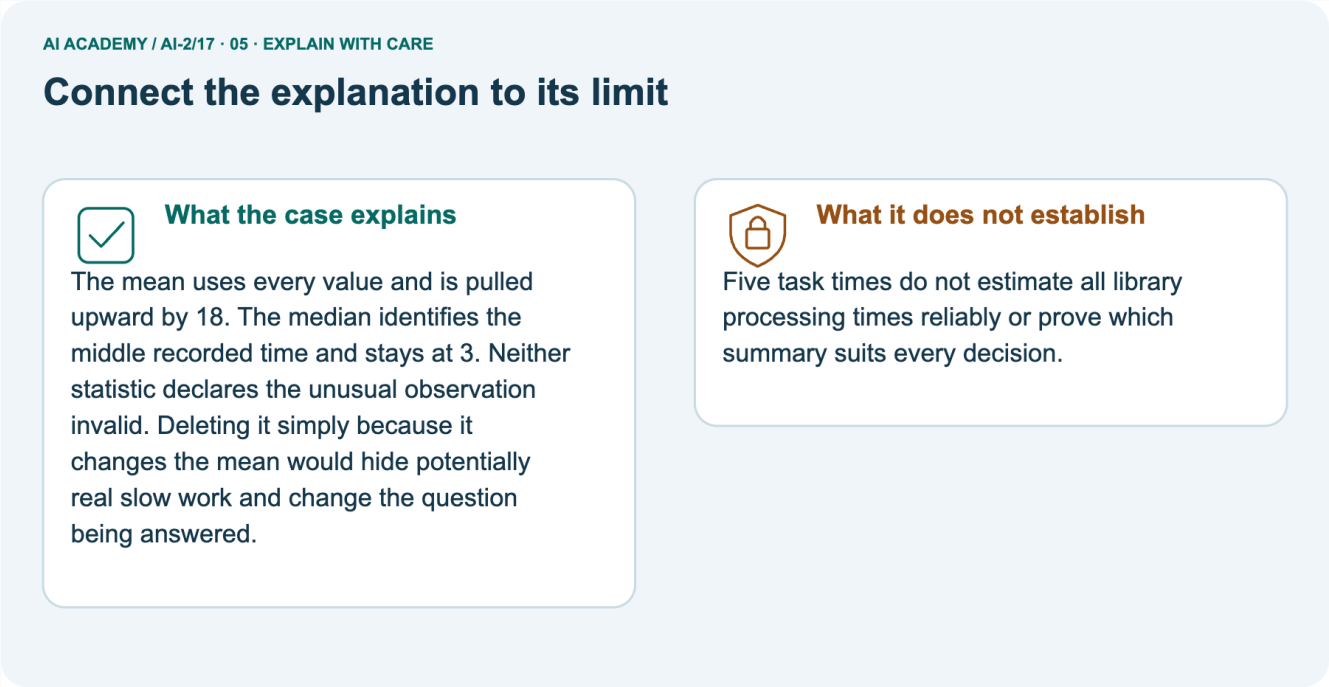
Original case: Change a value only if **evidence** supports a correction; preserve the raw record. Changed condition: An authorised log confirms 18 was a transcription error for 3.

Expected result: Corrected **mean** is 3 minutes and **median** is 3. Reason: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **mean** uses every value and is pulled upward by 18. The **median** identifies the middle recorded time and stays at 3. Neither statistic declares the unusual **observation** invalid.

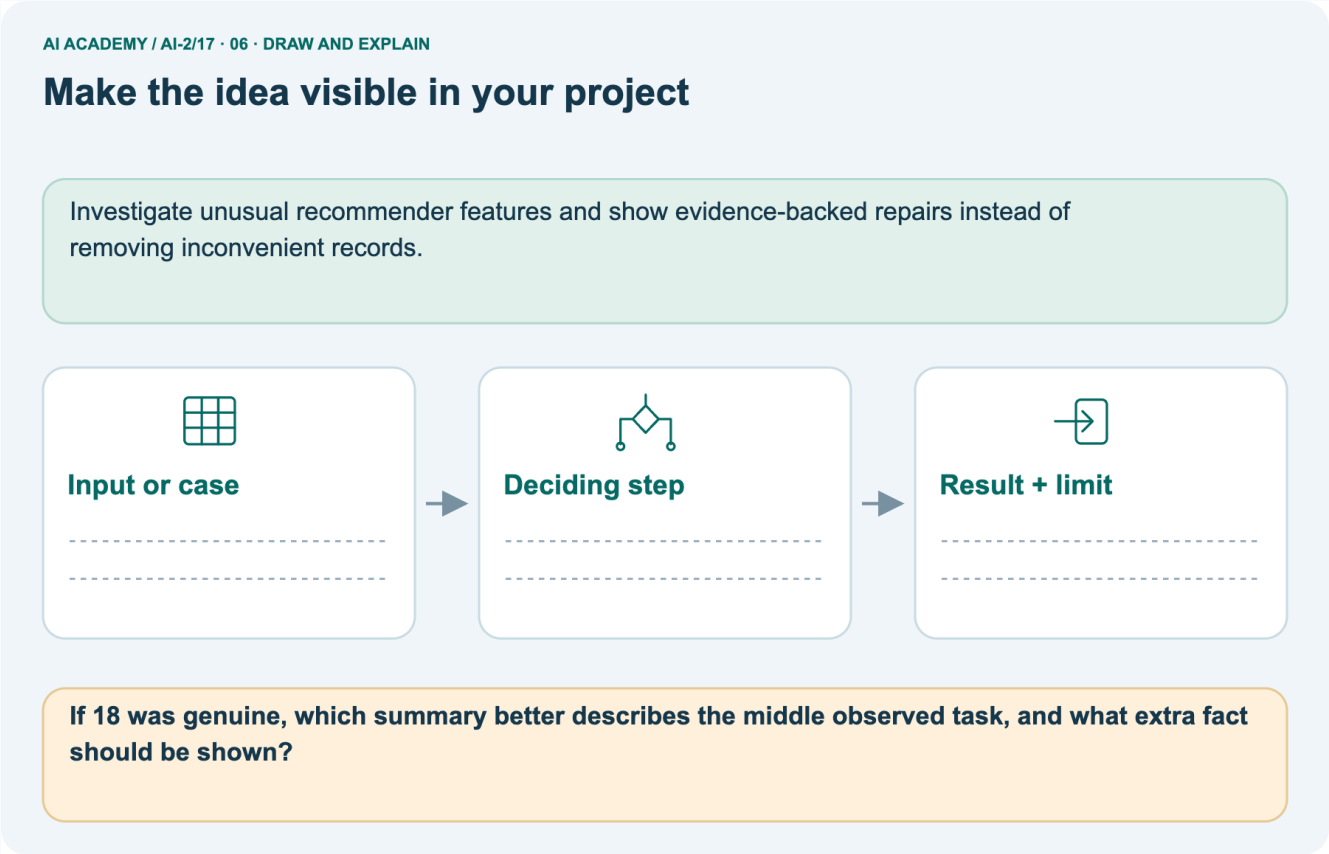
Deleting it simply because it changes the **mean** would hide potentially real slow work and change the question being answered.

Limit: Five task times do not estimate all library **process**ing times reliably or prove which summary suits every decision.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Investigate unusual recommender features and show **evidence**-backed repairs instead of removing inconvenient records. Draw your **input**, deciding step, result and limit.

Supported question: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Noise and Outliers: the case

02. Follow the deciding steps

03. Retain unusual recorded measurements

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.